KALYANI GOVERNMENT ENGINEERING COLLEGE
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B.Tech in Electrical Engineering

4-WAY TRAFFIC LIGHT CONTROL SYSTEM

WITH EMERGENCY VEHICLE OVERRIDE & PEDESTRIAN CROSSING LOGIC

A Project Report on Industrial Automation Using Siemens TIA Portal V20

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Programme	B.Tech in Electrical Engineering
Year / Semester	4th Year — 2025–2026
PLC Platform	Siemens S7-1200 (Simulated via PLCSIM)
Software Used	TIA Portal V20, WinCC Runtime Advanced
Date	May 2026

1. Abstract

This project presents the design, programming, and simulation of a 4-way traffic light control system using a Siemens S7-1200 Programmable Logic Controller (PLC) and Siemens TIA Portal V20. The system implements a step-counter based sequence control architecture using TON timers and MOVE instructions to cycle traffic signals across four road zones: North, South, East, and West.

The project incorporates three key advanced features: (1) an emergency vehicle override that forces all road signals to RED and freezes the sequence counter upon detection of an emergency input; (2) pedestrian crossing logic that latches button press requests, activates walk signals during the corresponding green phase, and extends the green phase duration to allow safe pedestrian crossing; and (3) a fully functional Human-Machine Interface (HMI) developed in Siemens WinCC Runtime Advanced, providing real-time visualization of traffic signal states, step counter, timer values, and emergency status.

The entire system was designed without physical hardware using Siemens PLCSIM for PLC simulation and WinCC Runtime for HMI simulation, demonstrating a complete software-defined industrial automation solution. The project reflects core competencies in PLC programming, sequence control, timer logic, priority interrupt systems, and HMI design — skills directly applicable to industrial automation, smart city infrastructure, and process control engineering.

2. Introduction

2.1 Background

Traffic management at road intersections is one of the most critical applications of industrial automation in urban infrastructure. Traditional manually operated signals have been replaced by automated control systems that ensure safe, efficient, and coordinated flow of vehicles and pedestrians. Programmable Logic Controllers (PLCs) are the backbone of such automation systems, offering reliable, real-time control with deterministic scan cycle execution.

The Siemens S7-1200 PLC family, programmed through TIA Portal, is one of the most widely deployed control platforms in industrial and infrastructure automation globally. Its support for Ladder Diagram (LAD) programming, function block architecture, TON timer instructions, and seamless HMI integration with WinCC makes it an ideal platform for implementing complex sequential control systems.

2.2 Motivation

This project was undertaken to demonstrate practical competency in Siemens TIA Portal V20 following the completion of the 'Learn Siemens PLC & HMI 2025: TIA Portal V20 Complete Course' (Udemy, BAZ Controls, May 2026). The traffic light control system was selected because it encompasses the full range of automation concepts — sequence control, timer logic, priority interrupts, I/O mapping, function block design, and HMI development — within a single, coherent real-world application.

Unlike simple on/off control or conveyor belt simulations, a traffic intersection system introduces genuine complexity through multi-road coordination, emergency priority handling, and pedestrian safety logic — making it a meaningful demonstration of engineering competence.

2.3 Objectives

The primary objectives of this project are:

1. Design and implement a 4-way traffic light sequence using step-counter logic on Siemens S7-1200 PLC.
2. Implement an emergency vehicle override system that immediately halts normal operation and forces all roads to a safe state.
3. Develop pedestrian crossing logic with request latching and green phase extension to ensure safe crossing time.
4. Create a fully functional HMI in WinCC Runtime Advanced to visualize system status in real time.
5. Validate the complete system through PLCSIM software simulation without requiring physical hardware.

3. Problem Statement

Design and implement a fully automated 4-way road intersection traffic light control system using a Siemens S7-1200 PLC that satisfies the following requirements:

3.1 Normal Operation

- The system shall control traffic lights for four road zones: North, South, East, and West.
- North and South roads shall share the same phase (both green simultaneously), and East and West roads shall share the same phase.
- The sequence shall cycle through green, yellow, and red phases with configurable timer durations.
- The system shall start only when a dedicated SYSTEM ON pushbutton is activated.

3.2 Emergency Override

- Upon detection of an emergency vehicle input signal, the system shall immediately force ALL road signals to RED.
- All pedestrian walk signals shall be deactivated during an emergency.
- The step sequence counter shall be frozen at its current value during an emergency.
- Upon clearing of the emergency signal, the system shall resume from the same step without resetting.

3.3 Pedestrian Crossing

- Each road pair (NS and EW) shall have a dedicated pedestrian pushbutton.
- A pedestrian request shall be latched and remembered even after the button is released.
- The walk signal shall only activate during the corresponding road's green phase.
- When a pedestrian request is active, the green phase shall be extended by an additional preset time to allow safe crossing.
- The pedestrian request latch shall be cleared automatically when the green phase ends.

3.4 HMI Requirements

- A graphical HMI shall display a top-down intersection view with animated traffic light indicators for all four roads.
- The HMI shall display real-time step counter value, timer elapsed time, and emergency status.
- SYSTEM ON and EMERGENCY SWITCH controls shall be operable from the HMI.
- Pedestrian request buttons shall be accessible from the HMI.

4. System Architecture

4.1 Hardware Platform

The project uses the following Siemens hardware platform (simulated via PLCSIM):

PLC Model	Siemens SIMATIC S7-1200 CPU 1212C DC/DC/RLY
Digital Inputs	8 x DI (I0.0 – I0.5, I1.0 – I1.1)
Digital Outputs	17 x DQ (Q0.0 – Q0.7, Q1.0 – Q1.7, Q2.0)
Programming SW	TIA Portal V20
Simulation	Siemens PLCSIM V20
HMI Panel (sim)	SIMATIC KTP700 Basic (WinCC Runtime Advanced)
HMI Software	WinCC Runtime Advanced V17

4.2 Software Architecture

The project is structured as a TIA Portal project with two Function Blocks (FB) called from the Main Organization Block (OB1):

- FB1 — I/O Mapping: Maps physical I/O addresses to named tags in the I/O Mapping Tags DB (DB1). This layer abstracts hardware addressing from the control logic.
- FB2 — Traffic Lights Control Main: Contains all control logic organized into 8 networks covering sequence control, zone outputs, pedestrian logic, and emergency handling.

4.3 Traffic Phase Sequence

The system operates on a 4-step sequence as follows:

Step	North/South	East/West	Duration	Description	Timer
1	GREEN	RED	11 seconds	NS green phase — NS vehicles move	T1
2	YELLOW	RED	4 seconds	NS yellow — NS vehicles prepare to stop	T2
3	RED	GREEN	11 seconds	EW green phase — EW vehicles move	T3
4	RED	YELLOW	4 seconds	EW yellow — EW vehicles prepare to stop	T4

After Step 4 completes, the sequence loops back to Step 1, creating a continuous cycle. The sequence is controlled by the STEP COUNTER (Word type) stored in the Traffic Lights Main Tags DB (DB5).

5. Components Used

5.1 Software Components

Component	Version	Purpose
Siemens TIA Portal	V20	PLC programming environment
Siemens PLCSIM	V20	Software PLC simulation
WinCC Runtime Advanced	V17	HMI design and runtime simulation
LAD (Ladder Diagram)	—	PLC programming language used

5.2 PLC Instruction Set Used

Instruction	Type	Usage in Project
MOVE	Data operation	Write step values (1–4) into STEP COUNTER
TON	Timer	Time each traffic phase (green 11s, yellow 4s, pedestrian 10s)
== (EQ)	Comparator	Check current STEP COUNTER value to control outputs and timers
SET (S)	Bit operation	Latch pedestrian request bits (NS_PED_REQ, EW_PED_REQ)
RST (R)	Bit operation	Clear pedestrian request latch when green phase ends
NO contact	Contact	Normal open — standard logic contacts
NC contact	Contact	Normal closed — emergency freeze, pedestrian hold
Output coil	Output	Drive physical output tags for lights

5.3 Tag Databases

Two Data Blocks (DBs) store all project tags:

DB1 — I/O Mapping Tags

Contains all physical I/O interface tags including all light outputs (RED/GREEN/YELLOW for all 4 roads), walk outputs, emergency input/output, and pedestrian button inputs.

DB5 — Traffic Lights Main Tags

Contains internal control tags: STEP COUNTER (Word), SYSTEM ON (Bool), NS_PED_REQ (Bool), EW_PED_REQ (Bool), PED COUNTER HOLD 1 (Bool), PED COUNTER HOLD 2 (Bool), NS_PED_TIMER (DInt), EW_PED_TIMER (DInt).

6. I/O Mapping

6.1 Digital Inputs

Address	Tag Name	I/O Mapping Tag	Function
%I0.0	PED_PB_NORTH	WALK_REQ_NS_INPUT	NS pedestrian button (North)
%I0.1	PED_PB_SOUTH	WALK_REQ_NS_INPUT	NS pedestrian button (South)
%I0.2	PED_PB_EAST	WALK_REQ_EW_INPUT	EW pedestrian button (East)
%I0.3	PED_PB_WEST	WALK_REQ_EW_INPUT	EW pedestrian button (West)
%I0.4	Emergency	EMERGENCY_INPUT	Emergency vehicle sensor
%I0.5	System_ON_PB	SYSTEM ON (DB5)	System start pushbutton

6.2 Digital Outputs

Address	Tag Name	I/O Mapping Tag	Function
%Q0.0	RED_NORTH	RED_NORTH_OUTPUT	North road red light
%Q0.1	GREEN_NORTH	GREEN_NORTH_OUTPUT	North road green light
%Q0.2	YELLOW_NORTH	YELLOW_NORTH_OUTPUT	North road yellow light
%Q0.3	WALK_NORTH	WALK_NS_OUTPUT	North pedestrian walk signal
%Q0.4	RED_SOUTH	RED_SOUTH_OUTPUT	South road red light
%Q0.5	GREEN_SOUTH	GREEN_SOUTH_OUTPUT	South road green light
%Q0.6	YELLOW_SOUTH	YELLOW_SOUTH_OUTPUT	South road yellow light
%Q0.7	WALK_SOUTH	WALK_NS_OUTPUT	South pedestrian walk signal
%Q1.0	RED_EAST	RED_EAST_OUTPUT	East road red light
%Q1.1	GREEN_EAST	GREEN_EAST_OUTPUT	East road green light
%Q1.2	YELLOW_EAST	YELLOW_EAST_OUTPUT	East road yellow light
%Q1.3	WALK_EAST	WALK_EW_OUTPUT	East pedestrian walk signal

Address	Tag Name	I/O Mapping Tag	Function
%Q1.4	RED_WEST	RED_WEST_OUTPUT	West road red light
%Q1.5	GREEN_WEST	GREEN_WEST_OUTPUT	West road green light
%Q1.6	YELLOW_WEST	YELLOW_WEST_OUTPUT	West road yellow light
%Q1.7	WALK_WEST	WALK_EW_OUTPUT	West pedestrian walk signal
%Q2.0	EMERGENCY LIGHT	EMERGENCY_OUTPUT	Emergency indicator light

7. Control Logic Design

7.1 Step Counter Mechanism

The entire sequencing logic is built around a single STEP COUNTER tag of type Word stored in DB5. The counter holds values 1 through 4, each representing a specific traffic phase. The PLC checks this value every scan cycle and activates the corresponding outputs.

The counter is initialized by a MOVE instruction that writes the value 1 into STEP COUNTER when SYSTEM ON is TRUE and STEP COUNTER equals 0 (default power-up state). This single initialization rung starts the entire sequence.

Each step has a dedicated TON timer. The timer's IN pin is driven by a comparator contact that checks if STEP COUNTER equals that step's number. An NC EMERGENCY_OUTPUT contact is wired in series with every timer's IN condition — when an emergency is active, this NC contact opens, the timer's IN pin goes FALSE, the timer freezes and resets, and the step cannot advance. The sequence is frozen at its current step.

When a timer's Q output goes TRUE (phase duration elapsed), a MOVE instruction writes the next step value into STEP COUNTER. The comparator condition for the current step immediately goes FALSE, the timer resets automatically, and the next step's timer begins — creating a self-sustaining automatic sequence loop.

7.2 Output Assignment Logic

Each traffic light output is controlled by a dedicated rung using a comparator contact on STEP COUNTER. For example, GREEN_NORTH_OUTPUT is driven by [STEP COUNTER == 1] AND [NC EMERGENCY_OUTPUT]. This means the green light is ON only during step 1 and is forced OFF during any emergency condition.

The red light outputs use a parallel structure — they are activated both during their normal red phase steps AND during emergency via a parallel NO EMERGENCY_OUTPUT contact. This ensures all roads show RED during an emergency regardless of the current step.

7.3 Emergency Override Logic

The emergency override operates at two levels:

- Timer freeze: NC EMERGENCY_OUTPUT contacts in series with all timer IN conditions prevent any step advancement during emergency.
- Output override: A parallel NO EMERGENCY_OUTPUT contact on every RED output rung forces all RED lights ON simultaneously when emergency is active.
- Walk signal cutoff: Emergency conditions prevent walk signals from activating, ensuring pedestrian safety during emergency vehicle passage.
- Seamless resume: Because the step counter is frozen (not reset), the sequence resumes from exactly the same phase when emergency clears — no disruptive restart.

7.4 Pedestrian Crossing Logic

The pedestrian logic operates through four mechanisms working together:

7.4.1 Request Latching

When a pedestrian pushbutton is pressed, a SET instruction writes TRUE into NS_PED_REQ (or EW_PED_REQ for East/West). This latch bit remains TRUE even after the button is released. A separate RST rung clears the latch when the step counter advances away from the green phase step (Step 2 for NS, Step 4 for EW).

7.4.2 Walk Signal Activation

The walk output (WALK_NS_OUTPUT) is driven by [NS_PED_REQ] AND [STEP COUNTER == 1]. This ensures the walk signal only activates when a request is pending AND the road is currently in its green phase — never during red or yellow phases.

7.4.3 Green Phase Extension

A PED COUNTER HOLD bit (Bool) controls whether the step can advance. When NS_PED_REQ is TRUE, PED COUNTER HOLD 1 becomes TRUE, and an NC contact of this bit is placed in series after T1.Q in the step advancement rung. This blocks the MOVE instruction from firing even after T1 has finished its normal duration.

A separate pedestrian extension timer (T5_NS_PED, PT=T#10S) runs when WALK_NS_OUTPUT is active. When this timer completes, its Q output drives a second parallel branch that also connects to the MOVE block — with T1.Q as a prerequisite (both T1 AND T5_NS_PED must be complete). Only when both timers have finished does the step advance.

7.4.4 Timer Linkage

The NS_PED_TIMER and EW_PED_TIMER DInt tags in DB5 store the elapsed time values from the pedestrian timers, which are displayed on the HMI to show crossing time remaining.

8. Program Network Structure

The project contains two Function Blocks with the following network organization:

FB1 — I/O Mapping (6 Networks)

Network	Name	Description
1	INPUTS	Maps physical input addresses to named I/O tags for buttons and emergency
2	OUTPUT_NORTH	Maps I/O tags to physical output addresses Q0.0–Q0.3 (North lights + walk)
3	OUTPUT_SOUTH	Maps I/O tags to physical output addresses Q0.4–Q0.7 (South lights + walk)
4	OUTPUT_EAST	Maps I/O tags to physical output addresses Q1.0–Q1.3 (East lights + walk)
5	OUTPUT_WEST	Maps I/O tags to physical output addresses Q1.4–Q1.7 (West lights + walk)
6	EMERGENCY_OUTPUT	Maps EMERGENCY_OUTPUT tag to physical address Q2.0

FB2 — Traffic Lights Control Main (8 Networks)

Network	Name	Description
1	LIGHTING SEQUENCE CIRCUIT	Step counter initialization, 4 TON timers with emergency freeze, 4 MOVE instructions for step advancement
2	NORTH ZONE	Output assignment rungs for North road: GREEN (step 1), YELLOW (step 2), RED (steps 3,4 + emergency)
3	SOUTH ZONE	Output assignment rungs for South road: GREEN (step 1), YELLOW (step 2), RED (steps 3,4 + emergency)
4	EAST ZONE	Output assignment rungs for East road: RED (steps 1,2), GREEN (step 3), YELLOW (step 4 + emergency)
5	WEST ZONE	Output assignment rungs for West road: RED (steps 1,2), GREEN (step 3), YELLOW (step 4 + emergency)
6	PEDESTRIAN NORTH/SOUTH	NS pedestrian SET/RST latch, walk output, extension timer T5_NS_PED, modified MOVE with hold logic
7	PEDESTRIAN EAST/WEST	EW pedestrian SET/RST latch, walk output, extension timer T6_EW_PED, modified MOVE with hold logic
8	EMERGENCY CONDITION	EMERGENCY_INPUT driving EMERGENCY_OUTPUT relay tag

9. HMI Design

9.1 HMI Platform

The Human-Machine Interface was designed using Siemens WinCC in TIA Portal and simulated using WinCC Runtime Advanced V17. The HMI panel selected is the SIMATIC KTP700 Basic, simulated in the RT Simulator environment.

9.2 Screen Layout

The HMI consists of a single main screen featuring:

- A top-down graphical representation of a 4-way intersection with road markings for North (N), South (S), East (E), and West (W).
- Three-indicator traffic light sets for each road showing RED, YELLOW, and GREEN states with dynamic color changes based on PLC output tags.
- Cyan-colored pedestrian walk signal indicators adjacent to each road's pedestrian request button.
- Numeric display for STEP COUNTER showing the current sequence step (1–4) in real time.
- Timer elapsed value displays (+X.XXX format) showing pedestrian timer progress for NS and EW zones.
- Real-time date and time display.
- TURN ON SYSTEM button linked to SYSTEM ON tag in DB5.
- EMERGENCY SWITCH button linked to EMERGENCY_INPUT tag — styled with hazard yellow-black pattern for visual prominence.
- Emergency warning triangle graphic (visible only when EMERGENCY_OUTPUT is TRUE).
- Siemens SIMATIC HMI branding frame for professional presentation.

9.3 HMI Tag Connections

All HMI elements are connected to tags in DB1 (I/O Mapping Tags) or DB5 (Traffic Lights Main Tags) with HMI/OPC UA access enabled. This allows the WinCC Runtime to read output states and write button commands directly to the PLC data blocks.

10. Simulation and Testing

10.1 Simulation Environment

The project was validated entirely in software using Siemens PLCSIM V20 for PLC simulation and WinCC Runtime Advanced for HMI simulation. PLCSIM creates a virtual S7-1200 CPU instance that executes the ladder logic at full scan cycle speed, allowing complete functional testing without physical hardware.

10.2 Test Scenarios

Test Scenario	Expected Result	Status
System startup — press SYSTEM ON	STEP COUNTER initializes to 1, North/South GREEN activates	PASS
Normal cycle — wait for T1 to complete	Step advances 1→2, North/South YELLOW activates for 4s	PASS
Full cycle — steps 1→2→3→4→1	Complete cycle with correct lights per zone at each step	PASS
Emergency trigger during step 1	All roads force RED, timers freeze, step counter holds at 1	PASS
Emergency clear	Sequence resumes from step 1 without reset	PASS
NS pedestrian button during step 1	Walk signal activates, green phase extends by 10s before advancing	PASS
NS pedestrian button during step 3 (wrong phase)	Walk signal does NOT activate — request ignored	PASS
Emergency trigger during pedestrian crossing	Walk signal turns OFF, all roads RED, timer frozen	PASS
EW pedestrian button during step 3	EW walk signal activates, step 3 green extends by 10s	PASS
HMI STEP COUNTER display	Numeric display updates in real time with each step change	PASS

11. Results and Observations

The 4-way traffic light control system was successfully designed, programmed, and validated in the Siemens TIA Portal V20 environment. All test scenarios passed without errors. The following key results were observed:

- The step-counter based sequence architecture proved reliable and easy to debug. The use of a single STEP COUNTER Word tag as the master control variable simplified the output assignment logic significantly.
- The NC EMERGENCY_OUTPUT contact mechanism for timer freezing worked as designed — triggering the emergency input immediately halted all timers, and clearing it resumed the sequence without any step loss or reset.
- The SET/RST latch mechanism for pedestrian requests correctly retained button press information across scan cycles. No missed button presses were observed during simulation testing.
- The PED COUNTER HOLD mechanism successfully extended the green phase duration when pedestrian requests were active. The sequence only advanced after both the main green timer AND the pedestrian extension timer had completed.
- The WinCC HMI provided clear, real-time visual feedback of all system states. The emergency warning triangle, colored traffic light indicators, step counter display, and timer values all updated correctly during simulation.
- The modular FB/DB architecture (FB1 for I/O mapping, FB2 for control logic, DB1 for I/O tags, DB5 for internal tags) demonstrated good software engineering practice and made the program easy to navigate and maintain.

11.1 Limitations

- The system uses a 2-road shared phase model (NS shares one phase, EW shares one phase) rather than 4 independent road control. This is appropriate for a standard 4-way intersection but would require modification for complex intersections with turn lanes.
- Timer durations (11s green, 4s yellow, 10s pedestrian extension) are fixed in the program. A more advanced implementation would use recipe management or HMI-configurable timer presets.
- The simulation does not account for sensor-based adaptive timing (e.g., extending green based on vehicle queue length), which would be the next step toward a production-grade intelligent traffic system.

12. Conclusion

This project successfully demonstrates the design and implementation of a complete industrial automation solution for 4-way traffic light control using Siemens S7-1200 PLC and TIA Portal V20. The system incorporates all key features of a real-world traffic management system including automated sequence control, emergency vehicle priority, and pedestrian safety logic.

The project demonstrates practical competency across the full automation engineering stack — from PLC hardware architecture and I/O mapping, through ladder logic programming with timers, comparators, SET/RST instructions, and parallel branch logic, to HMI design and real-time simulation validation. The use of Function Block architecture, separate tag databases, and systematic network organization reflects professional software engineering practices applicable in industrial environments.

The pedestrian logic, in particular, required understanding of latch mechanisms, timer coordination, and priority hold logic — concepts that form the foundation of more complex industrial sequential control systems such as manufacturing line automation, batch process control, and conveyor sorting systems.

This simulation-based project demonstrates that meaningful, industry-relevant automation systems can be fully designed and validated in software, providing a strong foundation for deployment on physical PLC hardware in real industrial or infrastructure applications.

13. References

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14. Appendix — Tag Reference

A. Traffic Lights Main Tags (DB5)

Tag Name	Data Type	Start Value	Description
STEP COUNTER	Word	16#0	Current sequence step (1–4)
SYSTEM ON	Bool	false	System run/stop control
I/O ON/OFF SWITCH	Bool	false	Testing switch for I/O simulation
NS_PED_REQ	Bool	false	NS pedestrian request latch
EW_PED_REQ	Bool	false	EW pedestrian request latch
PED COUNTER HOLD 1	Bool	false	NS green phase hold for pedestrian
PED COUNTER HOLD 2	Bool	false	EW green phase hold for pedestrian
NS_PED_TIMER	DInt	0	NS pedestrian timer elapsed value
EW_PED_TIMER	DInt	0	EW pedestrian timer elapsed value

B. Timer Summary

Timer	DB	Preset Time	Purpose
T1	DB4	T#11S	NS green phase duration (step 1)
T2	DB6	T#4S	NS yellow phase duration (step 2)

Timer	DB	Preset Time	Purpose
T3	DB7	T#11S	EW green phase duration (step 3)
T4	DB8	T#4S	EW yellow phase duration (step 4)
T5_NS_PED	DB9	T#10S	NS pedestrian crossing extension timer
T6_EW_PED	DB10	T#10S	EW pedestrian crossing extension timer